

Task A. GreenLoop - E-Waste Logistics Optimization

Dhaka's e-waste collection system contains collection zones, candidate transfer hubs, licensed recycling plants, collection vehicles, and field inspectors. Roads form an undirected network whose edge weights represent congestion-adjusted travel cost for the planning period. Your task is to process each GreenLoop scenario and produce an integrated logistics and enforcement plan.

The system first computes all-pairs shortest travel costs with Floyd-Warshall. It then selects a non-empty subset of at most K transfer hubs within the infrastructure budget, preferring the subset with the greatest potential formal-recycling throughput and then the lowest estimated hub-plus-transport cost. Kruskal's algorithm is used to obtain the minimum-cost connected backbone between the chosen hubs and the specified central plant. Edmonds-Karp max-flow determines how much e-waste can actually move through the zone $\rightarrow$ hub $\rightarrow$ plant network under hub and plant capacities.

For each vehicle, a multi-dimensional 0/1 knapsack chooses a high-value set of zone pickups without exceeding weight, volume, or hazardous-item capacity. An earliest-finish-time greedy scheduler keeps only pickups that fit the vehicle's daily availability. Finally, zones are stratified by illegal-dumping risk and sampled using randomized reservoir sampling; the sampled audits are then greedily scheduled across the available inspectors.

Input Format

The first line contains an integer T , the number of test cases. For each test case, the input is given in the following order:

Input item	Meaning
$N \ M$	number of road-network nodes and undirected roads
$u \ v \ c \ (M \text{ lines})$	road between nodes u and v with congestion-adjusted cost c
$Z \ P \ R$	number of collection zones, candidate hubs, and recycling plants
node $w \ vol \ h \ val \ risk \ s \ f \ (Z \text{ lines})$	zone node, waste weight, volume, hazard amount, recoverable value, dumping-risk score, pickup-window start, pickup-window finish
node fixed cap $(P \text{ lines})$	candidate hub node, fixed construction cost, and throughput capacity
node cap $(R \text{ lines})$	recycling-plant node and daily intake capacity
$K \ budget \ centralPlantIndex$	maximum hubs, infrastructure budget, and 1-based index of the central plant
V	number of collection vehicles
$wCap \ vCap \ hCap \ start \ finish \ (V \text{ lines})$	vehicle weight, volume, hazard capacities, and availability window
inspectors auditQuota seed	number of inspectors, number of zones requested for audit sampling, and random seed

Output Format

For each test case, print the following information. All hub, zone, vehicle, and plant identifiers shown in the output are 1-based.

```
Case x
Selected hubs: <hub indices>
Hub-selection potential throughput: <value>
Estimated hub+transport cost: <value>
Backbone MST cost: <value>
Maximum formal throughput: <value>
Vehicle i load value: <value>; picked zones: <indices>; scheduled zones: <indices>
... one vehicle line for each vehicle ...
Randomized audit sample: <zone indices>
Scheduled audits: <zone indices>
```

If the chosen backbone cannot be connected, print "Backbone MST: disconnected" instead of the MST-cost line. If no feasible non-empty hub subset satisfies the budget and connectivity conditions, print the case line followed by "No feasible hub subset under the given budget/connectivity constraints."

Selection and Scheduling Rules

- Hub selection: maximize potential throughput; if tied, minimize estimated hub + transport cost; if still tied, choose the lexicographically smallest hub-index list.
- Potential throughput is $\min(\text{total zone waste weight}, \text{total selected-hub capacity}, \text{total plant intake capacity})$.
- The backbone MST is built on the shortest-path metric between the selected hubs and the central plant.
- Only zones carrying positive max-flow are available for vehicle loading. Vehicles are processed in input order; zones actually scheduled for a vehicle are removed before the next vehicle is processed.
- Pickup scheduling uses earliest finish time. A zone is eligible only when its entire pickup window lies inside the vehicle availability window.
- Audit risk strata are: low risk < 40 , medium risk $40-69$, and high risk ≥ 70 . The given seed makes the randomized sampling reproducible.

Constraints / Assumptions

- Nodes are numbered from 1 to N , hubs from 1 to P , plants from 1 to R , and zones from 1 to Z .
- Road costs, construction costs, capacities, waste quantities, and pickup values are non-negative integers.
- $1 \leq P \leq 20$ and $1 \leq K \leq P$. The $P \leq 20$ limit comes from exhaustive candidate-hub subset enumeration in the supplied solver.
- $1 \leq \text{centralPlantIndex} \leq R$.
- $0 \leq \text{start} < \text{finish} \leq 24$ for zone and vehicle daily time windows.
- $0 \leq \text{auditQuota} \leq Z$. The random seed is read as an unsigned integer.
- Roads are treated as undirected. A pure MST guarantees minimum-cost connectivity, not single-edge-failure resilience.

Examples

Sample Input	Sample Output
<pre> 1 7 12 1 4 2 1 5 5 2 4 3 2 5 2 3 4 4 3 5 1 4 6 3 4 7 6 5 6 4 5 7 2 4 5 2 6 7 3 3 2 2 1 4 3 1 8 80 8 10 2 5 4 1 9 50 9 11 3 3 2 1 7 20 11 12 4 10 7 5 12 10 6 8 7 6 2 22 1 2 8 6 2 8 12 5 4 1 8 12 1 2 12345 </pre>	<pre> Case 1 Selected hubs: 1 2 Hub-selection potential throughput: 12 Estimated hub+transport cost: 71 Backbone MST cost: 5 Maximum formal throughput: 12 Vehicle 1 load value: 16; picked zones: 2 3; scheduled zones: 2 3 Vehicle 2 load value: 8; picked zones: 1; scheduled zones: 1 Randomized audit sample: 1 2 Scheduled audits: 1 </pre>

Sample Explanation

The best feasible hub subset is hubs 1 and 2. Their potential throughput is 12 units, and the estimated construction-plus-transport cost is 71. Kruskal connects the selected hubs and the central plant with an MST cost of 5, while Edmonds-Karp confirms that 12 units can reach the recycling plants. The knapsack and greedy stages assign zones 2 and 3 to Vehicle 1 and zone 1 to Vehicle 2. The seeded randomized audit stage samples zones 1 and 2, but the inspector schedule can accommodate only zone 1 without an interval conflict.